

BALAKUMARA M

Computer Science Engineering Student — Robotics, IoT & Embedded Systems
Thiruvanniyur, Chennai 600041 | +91 75502 16400 | bala2005.kmr@gmail.com
[LinkedIn](#) | [Portfolio](#)

OBJECTIVE

Computer Science Engineering student specializing in robotics, autonomous systems, IoT, and ServiceNow platform development. Experienced in embedded systems, automation, application development, and hardware-software integration. Skilled in Python, MySQL, ServiceNow, and UI/UX design, with hands-on experience building intelligent, scalable technology solutions.

SKILLS

- IoT development and hardware design; ServiceNow application development
- ArduPilot – Mission Planner
- Python Programming & Data Structures
- Databases: MySQL, Firebase
- Web: HTML, CSS, JavaScript
- UI/UX design: Figma, Adobe Illustrator
- Artificial Intelligence

EDUCATION

B.E., Computer Science Engineering — Sathyabama Institute of Science and Technology	CGPA: 9.25
Class XII — Velammal New Gen, Sholinganallur	78%
Class X — Velammal New Gen, Sholinganallur	79%

INTERSHIP EXPERIENCE

- Femtosoft Technologies** — Software Engineer Intern – 1 Month (03/06/2025 – 03/07/2025)
- Codoid Innovations** — UI/UX Designer Intern – 1 Month (03/12/2024 – 03/01/2025)

PROJECTS

- Navricon — Developed an autonomous robot capable of navigating and delivering small packages with minimal human intervention. Integrated sensors, motor control, navigation algorithms, and machine learning models for intelligent perception and decision-making, enabling accurate obstacle detection and autonomous navigation.
- Hydrocyclone Separation Unit — Designed an IoT and machine learning-based system to separate oil from water using a hydrocyclone mechanism. Integrated ESP32, sensors, and machine learning for oil detection/classification, enabling real-time monitoring and automated separation control.
- Campus Automated Leaf-Cleaning Rover — Built an autonomous rover to detect and collect dry leaves for smart campus maintenance. Used sensors and motor control for efficient navigation and cleaning.
- Grounded Policy Assistant — Developed a RAG-based policy assistant with hybrid BM25 and semantic retrieval for citation-backed answers. Implemented temporal reasoning to handle policy amendments and effective-date-based responses.
- Attendance App — Developed a Firebase-based attendance application with real-time data synchronization. Enabled secure attendance tracking and instant record updates.
- UI/UX App Interface — Designed a complete mobile application interface using Figma with a focus on user-friendly navigation. Created interactive prototypes following modern UI/UX principles.

CERTIFICATIONS

- ServiceNow Certified System Administrator (CSA) — ServiceNow
- ServiceNow Certified Application Developer (CAD) — ServiceNow
- NPTEL Certifications (3): Programming in Java, DBMS, Operating Systems (2023–2025)
- Cybernaut's UI/UX Course on Illustrator (2025)
- Robotics Workshop Conducted — Loyola Institute of Technology

ACHIEVEMENTS

- Designed Underwater Robotics project at Kurukshetra '25, Anna University
- Winner, Ideathon — Sathyabama Institute
- The Luminary Award, 2024–25 — Sathyabama Institute
- Selected for funding under Sathyabama Startup initiative

ADDITIONAL INFORMATION

- Languages known: English, Tamil | Professional traits: Time Management, Critical Thinking, Analytical Thinking, Communication, Teamwork, Adaptability, Problem-Solving